# What is the difference between a threat, vulnerability, risk and exploit? Give an example of each.

## Threat, Vulnerability, Risk & Exploit

| Concept | Simple Definition | Real-World Example |
|---|---|---|
| **Threat** | The potential danger or actor that wants to cause harm. | A ransomware group looking for a payday. |
| **Vulnerability** | A weakness, flaw, or bug in a system or its setup. | An unpatched, old version of a web server. |
| **Exploit** | The tool, script, or method used to take advantage of the flaw. | A public Python script that abuses that web server flaw. |
| **Risk** | The actual chance and business impact of the threat happening. | Losing money and customer data because the server got hacked. |